<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="4">Marks Available</th><th></th><th></th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>5</td><td>(a)</td><td>(i)</td><td>I</td><td>X^DY X^DX^d
X^D Y X^D X^d
All four correct = 2 marks
2/3 correct = 1 mark
0/1 correct = 0 marks</td><td></td><td>2</td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td>II</td><td><table><tr><td>Offspring genotypes</td><td>Offspring phenotype</td></tr><tr><td>X^DX^D</td><td>Normal female</td></tr><tr><td>$X^D X^d$</td><td>Normal female</td></tr><tr><td>X^DY</td><td>Normal male</td></tr><tr><td>$X^d Y$</td><td>X-SCID male</td></tr></table>
All correct = 2 marks
2/3 correct = 1 mark
0/1 correct = 0 marks
Accept ‘unaffected’</td><td></td><td>2</td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>64 cases = 2 marks
If incorrect award 1 mark for
<u>3139000</u>
48933
60 / 60.0 / 64.1 / 64.15 (incorrect sf)</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td></td></tr>
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Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		**Any four (x1) from:** A. folic acid (diet) produces more brown offspring / BPA (diet) produces more yellow offspring (1) B. Any correct reference to comparison of correct figures (1) C. there is methylation (of the promoter region of the gene) in brown mice / {no / less} methylation in yellow mice (1) D. Less mRNA in brown mice / mRNA only made during development in brown mice / more mRNA in yellow mice / mRNA produced through lifetime in yellow mice (1) E. methylation {silences / decreases expression of} the {agouti / yellow fur} gene / ORA (1) F. BPA de-methylates (the promoter region of) the gene / {without BPA / with folic acid} (the promoter region of) the gene is methylated (1)		1	3	4		2
		(ii)		• (So) {avoid / reduce} (addition of) BPA to (food) {plastics / containers} (1) • (Humans) exposed to (plastic) BPA can lead to (an increased incidence of) (diabetes / obesity} (1)			2	2		
				Question 5 total	**0**	**7**	**5**	**12**	**2**	**2**